



UE23CS342BA9 - Generative AI and its Applications

Data Resolution and Precision

Dr. UMA D

Professor

Department of Computer Science and Engineering

- What is Data Resolution?
 - Data resolution refers to the granularity and detail in the data used to train and fine tune models.
 - Higher resolution = more detail, but can lead to increased computational cost.
- What is Data Precision?
 - Data precision is the accuracy of the data values, ensuring minimal loss of information.
 - Precise data = more reliable outputs from the model.
- Relevance to LLMs:
 - LLMs require high quality data for effective fine tuning.

Why Data Resolution Matters?

- Data resolution refers to the level of detail and granularity in the input data used for training or fine tuning models.
- Higher resolution data ensures that models learn subtle features and hidden patterns, which significantly improves their performance.

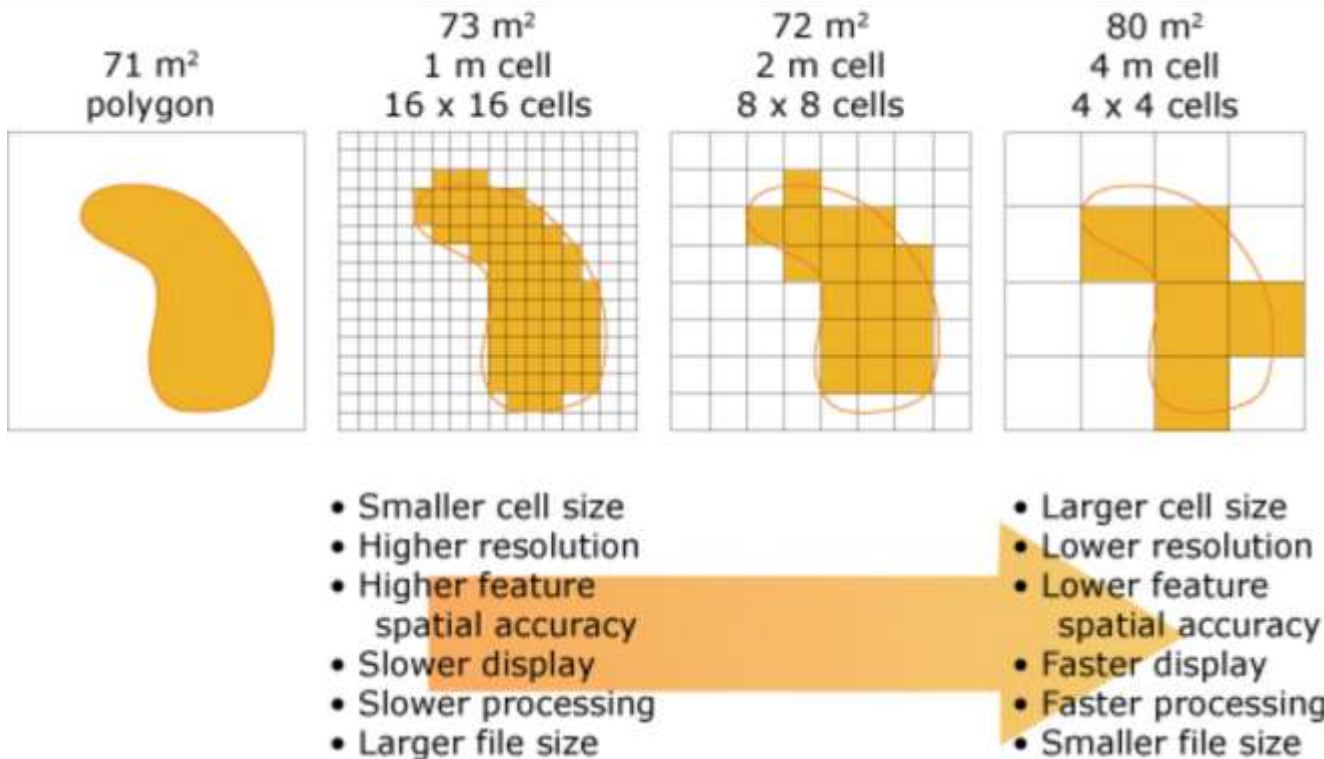
High Resolution Data

Captures fine details and subtle patterns critical for understanding complex tasks.

- Enables models to produce more accurate and nuanced outputs because they can better interpret the relationships and context within the data.

Benefits of High Resolution Data:

Improved accuracy in text generation and classification tasks



Why Data Resolution Matters?

- Enhanced ability to understand context dependent tasks (like sentiment analysis, language generation).
- Examples:
 - Text Data Resolution:
 - High Resolution: Full sentences with proper grammar, punctuation, and context. Example: "The restaurant had a cozy atmosphere and the food was absolutely delicious, especially the pasta."
 - Low Resolution: Isolated keywords or fragments. Example: "Restaurant good pasta cozy food."
 - Loss of context reduces the model's ability to differentiate between positive and neutral sentiments.

Why Data Resolution Matters?

- Image Data Resolution (in multimodal models):
 - High resolution images allow models to understand fine grained details (like facial expressions, object edges).
 - Low resolution images may cause blurring or loss of critical information.
- Numerical Data Resolution:
 - In tasks like time series prediction or financial forecasting, higher resolution data (like per second vs. per day observations) allows models to capture micro fluctuations and trends.



Scale 1:20,000
Cell size: 15 m



Scale 1:20,000
Cell size: 15.24 cm

Why Data Resolution Matters?

Low Resolution Data

- May lose critical details and lead to:
 - Generalized outputs: The model becomes less responsive to subtle nuances.
 - Ambiguity: Insufficient resolution reduces the clarity of the input signal, confusing the model.
 - Bias: Lower resolution data may reinforce incorrect patterns due to missing context.

Impact of Low Resolution Data:

Image Data:

Low resolution causes blurring, making it difficult for models to detect small but important features / like identifying a crack in a

Why Data Resolution Matters?

Impact of Low Resolution Data:

- Language Models:

- Missing sentence structure or punctuation can change the meaning.

- Example:

- "Let's eat, Grandma!" (invitation) vs. "Let's eat Grandma!" (ambiguity).

- In low resolution data, such distinctions may be lost, degrading the model's performance.

Time Series Models:

Low resolution data (like data sampled at weekly intervals instead of hourly) may miss short term trends, leading to inaccurate

Why Balancing Resolution Matters?

- While high resolution data offers better performance, it also comes with trade offs:
 - Higher Computational Cost: Training with high resolution data requires more memory and compute power.
 - Longer Training Times: Increased data detail results in longer processing and learning times.

Takeaways:

- Use high resolution data for tasks requiring fine grained understanding (like sentiment analysis, medical imaging).
- Balance resolution to optimize performance, cost, and training efficiency.

Always evaluate how resolution impacts model outputs based on

Data Precision in Fine Tuning

- Precision in Data Values:
 - Refers to how exactly data is represented (like decimal places, fixed point precision).
- Importance for LLMs:
 - Fine tuning LLMs with precise datasets leads to better learning.
 - Examples:
 - Text: Correct spelling, grammar, and word choice.
 - Numerical data: High precision values for tasks like prediction or regression.

Precision in Data Values

Precision refers to the exactness or granularity of the data used for training or fine tuning.

This can involve:



Text Data:

- Correct spelling, grammar, and word usage.
- Avoidance of ambiguous or contextually incorrect phrases.

Example:

- High precision: “Artificial intelligence enhances productivity by automating repetitive tasks.”
- Low precision: “AI is gud 4 automating.”

Numerical Data:

■ Use of accurate decimal values in datasets, especially for predictions or regressions.

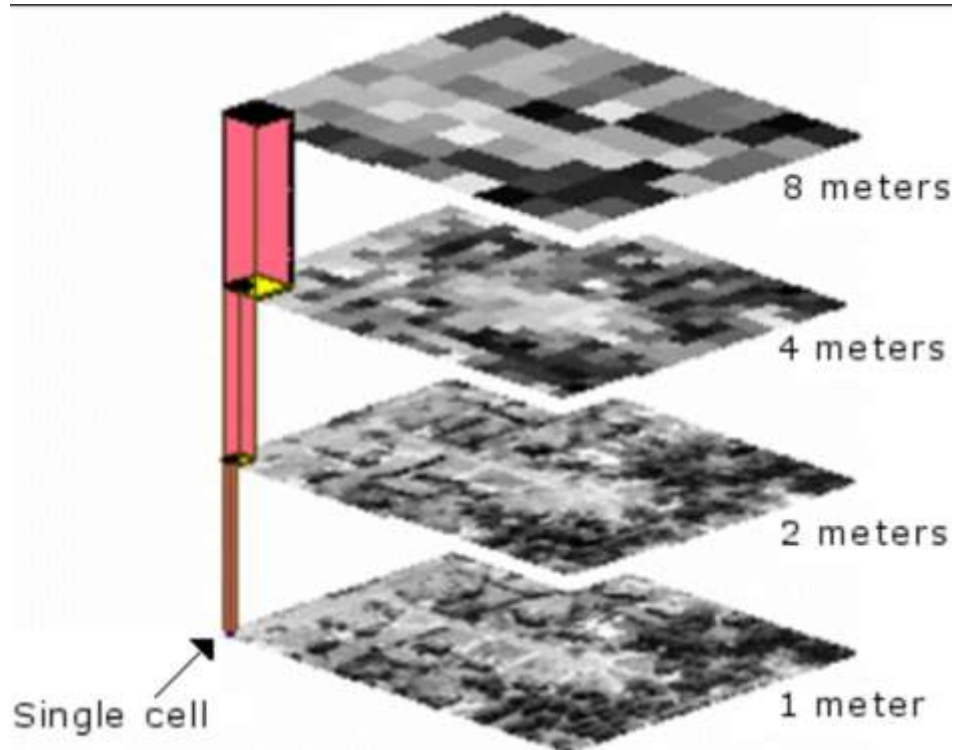
Example:

■ High precision: Sales data values like \$123.45 instead of rounded \$120 Low

Why Precision Matters in Fine Tuning LLMs?

- Models require precise data to learn relationships and patterns effectively.
- Benefits of precise data in fine tuning:
 - Accuracy:
 - Leads to better predictions and text generation.
 - Context Preservation:
 - Ensures the model understands and generates responses contextually.
 - Example: A high precision dataset distinguishing “bank” (financial institution) vs. “bank” (riverbank).
 - Error Reduction:
 - Fewer errors in downstream tasks, such as classification or translation.
 - Improved Generalization:

Fine tuned models on precise datasets can perform well on unseen data



Real World Examples of Precision in Fine Tuning

- Text Data:

- Training a chatbot for customer service:

- High precision ensures grammatically correct and professional responses.

- Example:

- Precise: “Your refund will be processed within 5 7 business days.”

- Imprecise: “Refund soon, wait 5 days.”

- Numerical Data:

- Predicting stock prices or weather conditions:

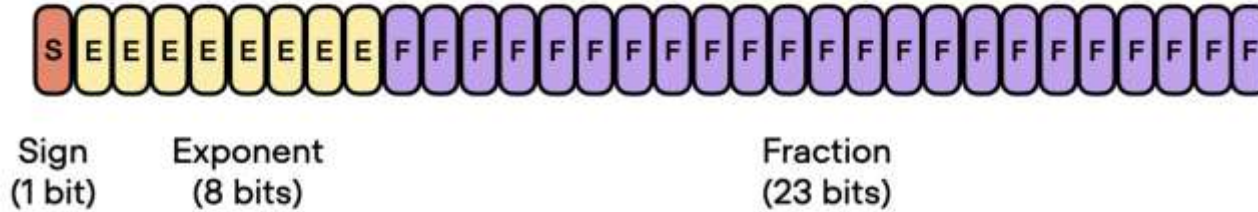
- High precision ensures accurate forecasts, like predicting rainfall as

Impact of Low Data Precision

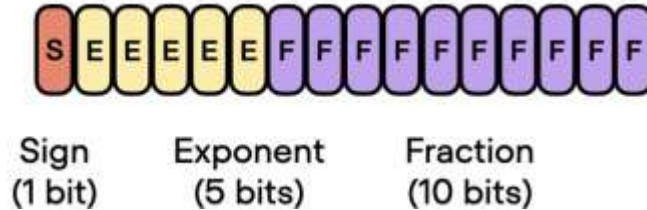
Loss of Information

- Reduced precision can lead to:
 - a. Ambiguity in Text Data:
 - Imprecise spelling, grammar, or context can confuse models.
Example: "There going to the store" (incorrect grammar) vs. "They're going to the store" (correct).
 - b. Misinterpretation of Numerical Data:
 - Rounded or coarse numerical data may obscure important trends or patterns.
Example:
Using rounded sales data like \$100,000 hides fluctuations between \$100.499 and \$99.501.

float 32



float 16 ("half" precision)



What are Precision Formats?

- Numerical formats used to represent data in computations, particularly for machine learning tasks.

■ Key factors:

 Floating-Point (fp): Handles real numbers with a wide range of values

- using a fractional and exponent part.

 Integer (int): Fixed-precision format for discrete numbers.

Why Precision Matters?

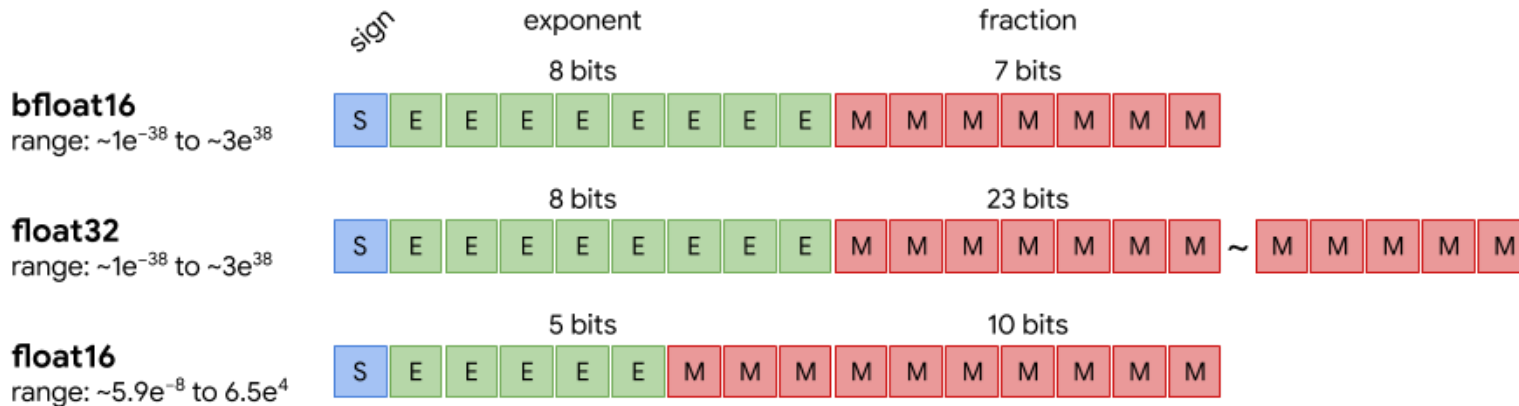
- Impacts accuracy, memory usage, and compute performance in models.

Trade-off between precision and efficiency.

Common formats:

- fp32 (32-bit float)

- fp16 (16-bit float)



Real World Impact of Low Precision

1. Text Data Example:

- Training an LLM on imprecise customer review data:
 - Imprecise: “The product is ok, I guess, maybe, not sure.”
 - High Precision: “The product is durable but lacks ease of use.”

Impact: Imprecise data leads to vague or irrelevant outputs, reducing the model's usefulness.

2. Numerical Data Example:

- Training a financial prediction model:
 - Imprecise: Stock prices rounded to the nearest \$10 (like \$100, \$110).
 - Precise: Stock prices recorded to two decimal places (like \$100.12, \$110.34).

How Low Precision Affects Tasks

- Sentiment Analysis:

- Imprecise reviews may lead to misclassification. Example:

- Imprecise: “Kinda good I think” vs. Precise: “The product is good but overpriced.”

- Classification Tasks:

- Imprecise labels or features can lead to misclassification, reducing model accuracy.

Consequences of Low Precision

1. Reduced Model Performance:

- Outputs become less reliable or contextually incorrect.

2. Increased Bias:

- Imprecision in training data may amplify unwanted biases, especially in diverse

- datasets.

3. Diminished Generalization:

Strategies for Increasing Data Resolution and Precision

1. Data Preprocessing

Proper preprocessing ensures that input data is clean, standardized, and ready for training. Key steps include:

Cleaning:

- Identify and remove irrelevant or noisy data that could confuse the model.

Examples:

- Removing typos, duplicates, and incomplete entries in text datasets.
- Filtering irrelevant rows in numerical datasets, like outliers that skew averages.

Tools: Libraries like Pandas, NumPy, or OpenRefine for structured data; NLTK or SpaCy for text cleaning.

- Normalization:

- Standardize numerical data to a fixed range (like between 0 and 1) or a standard distribution (like z scores).

- Importance:

- Ensures all features have equal weight in training, preventing dominance of large scale features.

- Example: Converting income data from \$10,000–\$100,000 to a range of 0–1 for training.

- Tools: Scikit learn's StandardScaler or MinMaxScaler.

Tokenization:

Break text into smaller, meaningful units (like words, subwords, or characters) to allow better data resolution for text based models.

- Techniques:

- Word level tokenization for simpler tasks.
- Subword level tokenization (like Byte Pair Encoding) for handling rare words.

Example: "unhappiness" → ['un', 'happiness'] with subword

- tokenization.

Tools: HuggingFace's tokenizers, SpaCy, or NLTK.

2. Data Augmentation

This technique artificially increases the size and diversity of training data, improving resolution and generalization:

For Text Data:

- Paraphrasing: Rewriting sentences to maintain the same meaning with different structure

- Synonym Replacement: Replacing words with synonyms without altering meaning.

- Example: "happy" → "joyful."

- Back translation: Translating text to another language and back to the original.

- Example: English → French → English.

For Numerical Data:

- Generate interpolated data points or perturb existing data slightly to increase diversity.

- Example: Adding Gaussian noise to numerical data to create slight variations.

For Images (if used in multimodal models):

- Rotate, crop, or flip images to increase variety in training data.

3. Precision Handling

- Maintaining or enhancing precision in datasets ensures that the model learns from accurate and reliable inputs:

- Higher Precision Formats:

- Use formats like float32 or float64 for numerical data to avoid rounding errors.

- Example: For financial data, precise values like \$123.45 instead of \$123

- improve model accuracy.

- Text Level Precision:

- Ensure proper spelling, grammar, and word choices in text datasets.

- Example: Correcting "ther are two cat" → "There are two cats."

- Dataset Curation:

- Choose or create datasets from trusted sources to maximize data reliability

Trade offs in Data Resolution and Precision

Higher Resolution & Precision

- Pros:
 - More Accurate Models:
 - Captures fine details and nuances that improve model performance.
 - Example: High resolution text data enables better sentiment analysis by recognizing subtle emotional cues.
 - Better Context Handling:
 - Precise datasets allow models to understand complex relationships and patterns.
 - Example: Numerical time series data with high precision captures micro trends in financial forecasting.

- Improved Generalization:
 - Models trained on high resolution data perform better on unseen examples.
- Cons:
 - Increased Computational Costs:
 - High resolution data requires more memory and computational power for training.
 - Example: Training a model on 4K image data vs. 480p requires significantly more resources.
 - Slower Training Times:
 - Processing high detail data takes longer, potentially delaying deployment.

Lower Resolution & Precision

- Pros:

- Faster Training:

- Simplified data reduces the processing time required for training.

- Example: Using sentence level data instead of paragraph level data for text classification.

-

- Reduced Resource Requirements:

- Lower memory and compute power are sufficient for coarse data.

- Example: Training on low resolution images or rounded numerical data

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- reduces hardware costs.

- Cons:
 -

- Loss of Important Information:

- Critical details may be missed, leading to subpar model performance

- Example: Rounded numerical data may hide subtle patterns (like small price fluctuations in stock market data).
- Poorer Model Quality:
 - Outputs become less nuanced and may lack depth.
 - Example: A chatbot trained on coarse text might produce generic and less helpful responses.

- The slides/diagrams in this PPT are an **adaptation**, **combination**, and **enhancement** of material from the following resources:
 1. <https://lightning.ai/pages/community/tutorial/accelerating-large-language-models-with-mixed-precision-techniques/>
 2. <https://moocaholic.medium.com/fp64-fp32-fp16-bfloat16-tf32-and-other-members-of-the-zoo-a1ca7897d407>
 3. <https://desktop.arcgis.com/en/arcmap/latest/manag>



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Generative AI and
its Applications

Dr. Uma D